Aidan Hirsch

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EDUCATION

Purdue University — 3.5/4.0 GPA – May 2028
Bachelor of Science in Biomedical Engineering — *West Lafayette, Indiana*

RESEARCH EXPERIENCE

Research Assistant — Jan 2026 – Present
Dadarlat Lab – Purdue University — *West Lafayette, IN*

- Modeled and labeled forelimb movements in mouse behavioral videos for DeepLabCut pose estimation model to be used in conjuction with two-photon microscopy neural data for motor cortex analysis
- Rebuilt and maintained lab's DeepLabCut training and analysis pipeline, restoring video pose estimation for multiple active experiments
- Trained electrode-implanted mice on somatosensory cortex behavioral tasks for neural stimulation studies

Research Assistant — Jan 2025 – Dec 2025
Neuroprostheses Research Lab – Purdue University — *West Lafayette, IN*

- Discovered novel correlation between modeled circuit values and electrode degradation by analyzing in-vivo electrical component parameters through UMAP statistical analysis
- Analyzed ultramicroelectrode array failure modes using 16-week electrochemical impedance spectroscopy study in rat somatosensory cortex
- Engineered reusable automation pipeline for data pre-processing, measurement modeling, and file preparation, currently saving over 8 months of manual labor and enabling rapid model iteration

Research Assistant — Apr 2021 – May 2024
Friel Lab – Burke Neurological Institute — *White Plains, NY*

- Designed quantitative bimanual hand assessment for children with cerebral palsy, boosting joint location accuracy by 30-50% over prior qualitative methods
- Engineered a custom drawer for rehabilitation hand assessment using Raspberry Pi and 3D cameras to capture participant joint locations in 3D space
- Developed Python program to interact with drawer assessment by determining when a participant moved or touched parts of the drawer through measurement of electrostatic discharge

LEADERSHIP AND PROJECTS

Chair — Jan 2026 – Present
Purdue IEEE Systems Man and Cybernetics Committee — *West Lafayette, IN*

- Leading Purdue's only neurotech club with 50+ active members in development of 128-channel EEG, concussion detection scoping research and publications, and international neural decoding competitions
- Teaching various aspects of BCI technology, including ICMS, microelectrode arrays, neurotech market developments, neural anatomy, and computational neuroscience on a weekly basis to club members
- Spearheading PCB design workshops for BME and ECE undergraduates to learn how to design PCBs from schematic, to layout, to part selection, to completed board

EEG Build Team Lead — Aug 2025 – Present
Purdue IEEE Systems Man and Cybernetics Committee — *West Lafayette, IN*

- Directing Electrical team in engineering a custom 128-electrode EEG system from component level encompassing custom ADC and microcontroller board layouts
- Designed, routed, and fabricated multi-layer printed circuit boards to integrate 16 ADS1299 ADCs to 5 ESP32-S3 microcontrollers to achieve high SNR neural signal acquisition
- Implementing embedded systems network for ESP microcontrollers to synchronize and stream data from all 128 electrode channels

SKILLS

Engineering Tools: KiCad | LTSpice | DeepLabCut | Solidworks | Fusion
Data Analysis: Python (pandas, NumPy, matplotlib, seaborn) | CEBRA | MATLAB | R | C
Biomedical Techniques: ICMS | Impedance spectroscopy | Rodent handling (IACUC) | Cyclic Voltammetry